Nine Principles for Designing Living Machines

1. Microbial Communities:

The primary ecological foundations of living machines are predicated upon diverse microbial communities obtained from a wide range of aquatic (marine and freshwater) and terrestrial environments. In addition, organisms from chemically and thermally highly stressed environments are critical. Genetic engineering cannot do what constellations of natural organisms can accomplish when they work in concert.

2. Photosynthetic Communities:

Sunlight-powered photosynthesis is the primary driving force of these systems. Anaerobic phototropic microbes, cyanobacteria, algae, and higher plants must be linked in a dynamic balance with the heterotropic microbial communities.

3. Linked Ecosystems and the Law of the Minimum:

At least three distinct types of ecological systems need to be linked together to produce living machines that carry out self design and self repair through time. Such systems have the theoretical ability to span centuries and possibly millennia.

4. Pulsed Exchanges:

Nature works in short term/long term pulses which are both regular and irregular. This pulsing is a critical design force and helps maintain diversity and robustness. Pulses need to be intrinsic to design.

5. Nutrient and Micronutrient Reservoirs:

Carbon/Nitrogen/Phosphorus ratios need to be regulated and maintained. A full complement of macro and trace elements needs to be in the system so that complex food matrices can be established and allowed to "explore" a variety of successional strategies over time. This will support biological diversity.

6. Geological Diversity and Mineral Complexity:

Living machines can simulate a rapid ecological history by having within them minerals from a diversity of strata and ages. The geological materials can be incorporated into the sub-ecosystems relatively quickly by being introduced as ultra fine powders which can be solubilized over short time frames.

7. Steep Gradients:

Steep gradients are required within and between the sub-elements of the system. These include redox, pH, humic materials, and ligand or metal-based gradients. These gradients help develop the high efficiencies that have been predicted for living machines.

8. Phylogenetic Diversity:

In a well engineered ecosystem all phylogenetic levels from bacteria to vertebrates should be included. System regulators and internal designers are often unusual and unpredictable

organisms. The development of various phyla has arisen to a large extent from the strategic exploration of the total global system over a vast period of time. This time can be compressed with the consequences of this evolution.

9. The Microcosm as a Tiny Mirror Image of the Macrocosm:

This ancient hermetic law applies to ecological design and engineering. As much as possible, global design should be miniaturized in terms of gas, mineral, and biological cycles. The big system relationships need to be maintained in the living machine.